

NAWAL SHAHID

Full Stack Engineer | React.js • Node.js • MERN Stack • AI Integration
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SUMMARY

Full Stack Engineer with 3+ years of experience building and deploying web, mobile, and AI-powered applications using React, Node.js, MongoDB, and React Native. Delivered a production management system for a Tanzania-based textile manufacturer, RAG-powered document intelligence applications, LLM-integrated analytics platforms, and an accessibility-focused ed-tech platform. Experienced in REST API design, JWT authentication, role-based access control, MongoDB schema design, performance optimization, CI/CD, Firebase, and AI integration using Groq LLMs.

EXPERIENCE

Full Stack Engineer, Production Management System | [Textile Manufacturing Client](#) | Tanzania (Remote Contract) | 2026

- Designed and developed a production management system for textile manufacturing covering inventory tracking, order processing, production workflows, and financial reporting, built on a Node.js/Express.js REST API with JWT authentication and role-based authorization across Admin, Manager, and Operator roles.
- Designed MongoDB document schemas with embedded references and compound indexes to optimize query performance for reporting and inventory lookup across high-volume collections.
- Implemented end-to-end business workflows including procurement approvals, low-stock alerts, and automated report generation; deployed to production with environment-based configuration and structured error handling.
- Designed reusable RESTful API endpoints and input validation layers across inventory, production, procurement, and reporting modules, improving maintainability and reducing integration effort for future feature development.

Full Stack Web and Mobile Developer | [Kazim Trust, DigiLex Ed-Tech Platform](#) | Karachi, PK | 2024 – 2026

- Architected the complete MERN stack platform: Node.js/Express.js REST API with JWT authentication, RBAC, and MongoDB document schemas with compound indexes designed for efficient content retrieval and user progress queries.
- Built the React.js frontend with semantic HTML, keyboard navigation, and ARIA labels, achieving a Lighthouse accessibility score of 93; developed cross-platform React Native mobile app with Firebase Auth and Firestore offline persistence, reducing session drop-off by ~40%.
- Optimized application performance through lazy loading, route-based code splitting, offline persistence, and reusable React component architecture, improving usability on low-bandwidth connections.

Web Developer Apprentice | [AJD Marketing](#) | Qatar (Remote) | 2023 – 2024

- Built and maintained 4 client websites (React.js, Node.js, PHP, MySQL, Tailwind CSS), improving Lighthouse performance scores from 68 to 91 and reducing average page load time from 3.4s to 1.7s through Core Web Vitals tuning and SQL query optimization.

Open Source Contributor | [GitHub, Various Repositories](#) | Remote | 2026

- Resolved bugs, submitted pull requests, and collaborated with maintainers across open source repositories; contributed improvements to documentation, test coverage, and code quality within large existing codebases following Git branching workflows and code review standards.

PROJECTS

IntelliAI: AI-Powered Business Intelligence Platform | Streamlit | Groq Llama 3.3 70B | Plotly | Pandas | LLM Integration | Prompt Engineering

- Built an end-to-end self-service analytics platform with an 8-step workflow covering CSV/Excel ingestion, automatic encoding and delimiter detection, date parsing, data profiling, cleaning, EDA, and interactive Plotly dashboards with no coding required.
- Integrated Groq's Llama 3.3 70B via prompt-engineered pipelines for AI-generated business insights, anomaly detection, trend identification, conversational Q&A analytics, and automated report generation.
- Designed a modular data-processing pipeline supporting automated preprocessing, AI-assisted analysis, and interactive visualization, enabling non-technical business users to derive insights from raw data independently.

AI Study Brain: RAG Document Q&A Application | Python | Streamlit | FAISS | Sentence Transformers | Groq API | LangChain-style RAG

- Built a modular RAG pipeline: PDF upload, text extraction, configurable chunking, local sentence transformer embeddings, FAISS vector indexing, similarity search, and Groq API answer generation with source citations. Designed a scalable Retrieval-Augmented Generation architecture supporting semantic document search and context-aware question answering.

NexaPress: Full Stack CMS MongoDB | Express.js | React.js | Node.js | JWT | Firebase | Cloud Functions | Vercel

- Built two CMS architectures: a MERN version with JWT authentication, RBAC, MongoDB schema design, and dynamic API routing deployed on Vercel; and a serverless Firebase version with Firestore security rules, real-time sync, and Cloud Functions for automated publication workflows.

TECHNICAL SKILLS

Frontend: React.js, Next.js, JavaScript (ES6+), TypeScript, HTML5, Tailwind CSS, Material UI, Semantic HTML, ARIA, Accessibility

Backend and APIs: Node.js, Express.js, REST API Design, API Integration, JWT, RBAC, Serverless, Performance Optimization, System Design, Debugging

Databases: MongoDB, Mongoose, Firestore, PostgreSQL, MySQL, Schema Design, Indexing, Aggregation Pipelines

AI Engineering: Groq LLM, Retrieval-Augmented Generation (RAG), FAISS, Sentence Transformers, Prompt Engineering, Python, Pandas, Streamlit, Plotly

Mobile and DevOps: React Native, Expo, Kotlin, Firebase, Docker, GitHub Actions, CI/CD, Git, Code Review, Agile, Netlify, Vercel

EDUCATION

BSc Software Engineering | Jinnah University for Women, Karachi | 2022 – 2026 | CGPA: 3.5 / 4.0

Technical Management Co-Lead, Google Developer Student Club (GDSC) | Creative Art Manager, University Event Branding